Reading a Data Centre's Power, Water and Heat Numbers

A four-page guide for municipal staff and councils

Prepared by NextGenergy Technology Inc., Toronto. Version 1, September 2026. Free to copy and adapt. Corrections welcome: jim.li@nextgenergy.ai

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Why this guide exists

On 3 September 2026 the Government of Canada and the Federation of Canadian Municipalities published the Responsible Data Centre Development Principles. Two of the five principles put numbers on the table:

- **Principle 3** asks projects to prioritize "closed-loop or high efficiency water technologies, waste heat recovery, and low-emission energy sources," with water consumption and environmental impacts "measured and transparently reported using recognized standards."
- **Principle 4** asks proponents for "clear, project-appropriate and independently verifiable information on expected local impacts, including power and water use, infrastructure requirements, sound and emissions."

The principles are voluntary. What makes them real is the evidence a municipality asks for, and whether staff can read it. This guide covers the three numbers that decide most of a data centre's local impact, what each one hides, and the questions that get a usable answer.

We build and commission liquid cooling for data centres. We are not neutral about the technology, but we are neutral about who builds in your municipality. Everything below applies to any proponent.

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1. Power

The two loads. A data centre draws power for two things: the computers (the IT load) and everything that keeps them running, mainly cooling (the facility load). Proponents usually quote the IT load, because it is the number their customers buy. The grid sees the sum.

PUE. Power Usage Effectiveness is total facility energy divided by IT energy. A PUE of 1.2 means 20 percent overhead. Three things to know:

- A **design** PUE is a promise. A **measured** PUE is a fact. Ask which one you are being shown.
- PUE is an **annual** figure. A cold-climate site will look excellent in January and worse in July. Ask for monthly values or the annual average with the year stated.
- PUE says nothing about the IT load itself. A site can have a superb PUE and still draw 200 MW.

Who pays for the connection. Principle 2 says proponents pay for "new generation, transmission, substations and grid upgrades directly attributable to their development." The document that proves this is the utility's connection cost allocation, not the proponent's statement. Ask to see it.

Ask for:

Question	Number	Unit
Peak electrical demand at the point of connection		MW
Annual energy at the point of connection		MWh per year
IT load at full build		MW
PUE, measured or design, with the year and boundary stated		ratio
Connection and upgrade costs, and who pays each line		CAD, by party
On-site generation, fuel, and hours per year it is expected to run		MW, fuel, hours

2. Water

Where the water goes. Most data-centre water is used to reject heat. Evaporative cooling towers turn water into vapour and can consume millions of litres a day. Closed-loop systems, which Principle 3 favours, reject heat through dry coolers or radiators and consume far less. But "closed-loop" is not "zero":

- Closed loops still take **make-up water** for leaks, sampling and periodic fluid replacement.
- Some closed-loop designs add **evaporative assist** on the hottest days. Ask how many hours a year, and how much water on the peak day.
- Dry coolers trade water for **electricity**. A low water number can push the power number up. Ask for both together.
- Humidification and domestic use are small but not zero.

WUE. Water Usage Effectiveness is litres of water consumed per kilowatt-hour of IT energy. It is defined in ISO/IEC 30134-9. A fully closed-loop site reports a WUE near zero; an evaporative site can be above 1.5 L/kWh. As with PUE, ask whether the number is design or measured, and for which year.

Ask for:

Question	Number	Unit
Annual water withdrawal, by source (municipal, ground, surface)		m^3 per year
Peak-day withdrawal		m^3 per day
Annual water consumed (not returned)		m^3 per year
WUE, design or measured, with the year stated		L per kWh of IT
Discharge volume, temperature and treatment		m^3 per year, °C
Days per year the evaporative assist, if any, is expected to run		days

3. Heat

Every watt becomes heat. A 100 MW data centre releases close to 100 MW of heat, continuously, all year. Principle 3 names waste heat recovery. Here is what to know before you count it as a benefit.

Heat needs a receiver. Heat only has value if someone next door can use it: a district heating network, a greenhouse, a pool, an industrial process. Without a named receiver, the heat goes to the air, whatever the proponent's intentions. The question is not "will you recover heat" but "who will take it, and how much, in July."

Temperature decides who can take it. Air-cooled sites produce heat at 25 to 35 °C, which almost no one can use without a heat pump. Liquid-cooled sites with warm-water loops can return heat at 45 to 60 °C, which district heating and greenhouses can use with little or no lift. Ask for the temperature at the fence, not at the chip.

Two numbers, not one. Capacity in MW is what could be delivered on the best hour. Delivery in MWh per year is what a receiver actually takes. For reference, the best-known Canadian-climate example, Google's Hamina site in Finland, delivers about 40 GWh a year through a 7.5 MW heat-pump plant, which covers about 80 percent of a town of 20,000. That number is set by the town's demand, not by the site's output. Expect the same in your municipality.

Ask for:

Question	Number	Unit
Heat available at the site boundary, and at what temperature		MW at °C
Named receiver, and its annual heat demand		name, MWh per year
Heat expected to be delivered in year 1, 3 and 5		MWh per year
Who funds the connection, who operates it, who provides backup heat		party
What happens to the heat in months the receiver cannot take it		method
Energy Reuse Factor, if reported (ISO/IEC 30134-6)		ratio

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4. Method: who measured, and who signed

A number without its method is an opinion. Principle 4 asks for "independently verifiable" information. Four things make a number verifiable:

1. **The boundary.** Where was it measured? A PUE measured at the utility meter and one measured inside the building are different numbers. Ask for a single-line diagram with the metering points marked.
2. **The condition.** At what load, in which month, over what period? A cooling system tested at 30 percent load in February proves little about August at full load.
3. **The standard.** PUE, WUE and ERF are defined in the ISO/IEC 30134 series. The European Union's Delegated Regulation 2024/1364 sets a reporting template for data centres above 500 kW that Canadian municipalities can borrow from freely.
4. **The signature.** Who measured it, and were they independent of the proponent? Commissioning agents, utilities and third-party verifiers can sign. A marketing department cannot.

Put it in the development agreement. The most useful sentence a municipality can write is: "The proponent shall report annually, in a public document, the quantities in Schedule X, measured at the boundaries shown in Drawing Y, by a party independent of the proponent." Everything else in this guide is Schedule X.

One-page checklist

#	Ask	Unit	Standard or reference	Who signs
1	Peak demand at the point of connection	MW	utility connection agreement	utility
2	Annual energy at the point of connection	MWh	utility metering	utility
3	PUE, measured, annual, boundary stated	ratio	ISO/IEC 30134-2	commissioning agent or verifier
4	Connection and upgrade cost allocation	CAD by party	utility cost allocation letter	utility
5	On-site generation: fuel, MW, hours per year	MW, hours	provincial permit	proponent, permit authority
6	Annual water withdrawal and consumption, by source	m^3	ISO/IEC 30134-9	proponent, verifier
7	Peak-day withdrawal	m^3 per day	design basis	engineer of record
8	WUE, measured, annual	L per kWh	ISO/IEC 30134-9	verifier
9	Heat available at the boundary, MW at °C	MW, °C	design basis	engineer of record
10	Named receiver and its annual demand	MWh	receiver's own data	receiver
11	Heat delivered, MWh per year, by year	MWh	heat meter at the boundary	receiver and proponent
12	Metering boundary drawing	drawing	single-line diagram	engineer of record
13	Independent verifier named	name	development agreement	municipality
14	Annual public report commitment	date	development agreement	proponent

About NextGenergy

NextGenergy Technology Inc. designs, prefabricates and commissions liquid-cooling infrastructure for AI data centres, from the facility water interface to the rack. We are a Nova Scotia corporation registered to operate in Ontario, with our principal place of business in Toronto. Our team has delivered warm-water liquid cooling at scale in Asia, including heat delivered to district heating, greenhouses and aquaculture, and we take part in Open Compute Project working groups on liquid-cooling standards and commissioning.

We wrote this guide because the questions in it are the ones we answer on every project, and we would rather municipalities asked them of everyone.

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